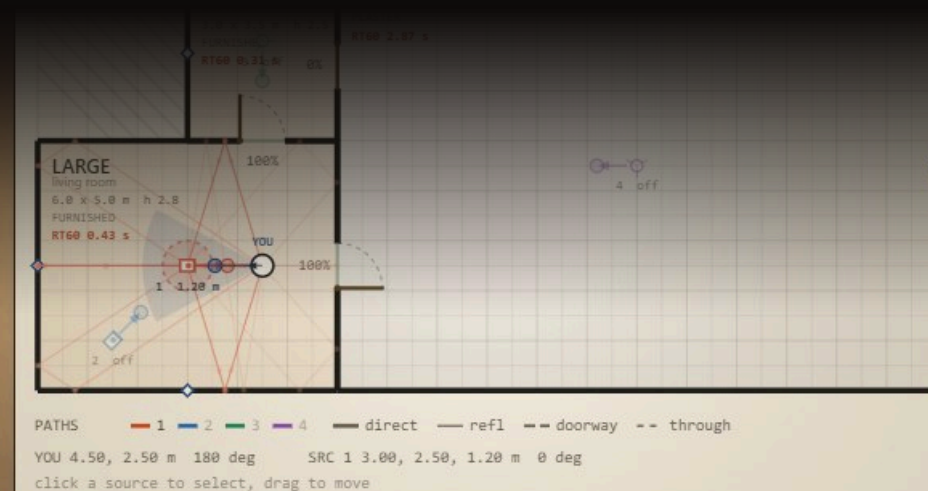


THIN WALLS

HANDBOOK

BUILD 260921.1



DOORS

S-L ☐ 100%

L-G ☐ 100%

S-G ☐ 0%

click a door in the plan to toggle

MATERIALS

	WALLS	FLOOR	CEILING
LARGE	FURNISHED	AS WALLS	AS WALLS
SMALL	FURNISHED	AS WALLS	AS WALLS
GIANT	PLASTER	AS WALLS	AS WALLS

floor and ceiling default to AS WALLS

BROKEN WALLS

LARGE W	☐	flat	☐	50%
LARGE S	☐	flat	☐	50%
SMALL W	☐	flat	☐	50%
SMALL N	☐	flat	☐	50%
SMALL E	☐	flat	☐	50%
SMALL W	☐	flat	☐	50%

push out to scatter · drag the marker in the plan

SOURCE

1 2 3 4

INPUT MAIN L+R

TYPE LOUDSPEAKER

HEIGHT 1.20 m

FACING 0 deg

DIRECT n/a

LEVEL +0.0 dB

x 3.00 m y 2.50 m

Three rooms, three doors, four sources and a head you can walk around with. Everything you hear is a path the engine has actually traced — through the air, off the walls, round the door frame or straight through the leaf.

September 2026.

AN APARTMENT YOU CAN PUT A SOUND INSIDE

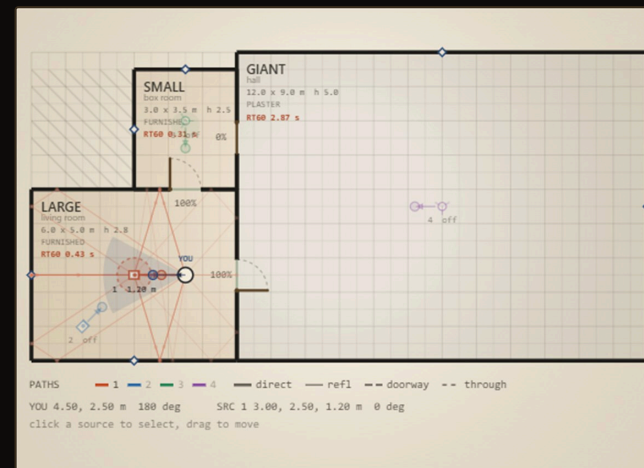
Thin Walls is a small flat: a **living room**, a **box room** and a **hall**, joined by three doors that you can open, shut or leave ajar. You place up to four sound sources in it, you stand in it yourself, and what comes out of the plugin is what your two ears would pick up from where you are standing and which way you are facing.

It is not a reverb with a room preset. Every arrival is computed as a journey: the straight line to each ear, the first and second bounces off the walls, the path that bends round a door frame, the muffled fraction that gets through a shut door or a party wall, and the tail each room builds for itself. Move the source a metre and the whole set of journeys changes, which is why close, far and next-door sound different in character and not merely in level.

There is no reverb-amount control, because the amount is decided by where you stood.

INSTALLING

Copy the whole **Thin Walls.vst3** folder into your VST3 folder — normally `C:\PROGRAM FILES\COMMON FILES\VST3` — and rescan in your DAW. The standalone in the same download needs no installing; double-click it.

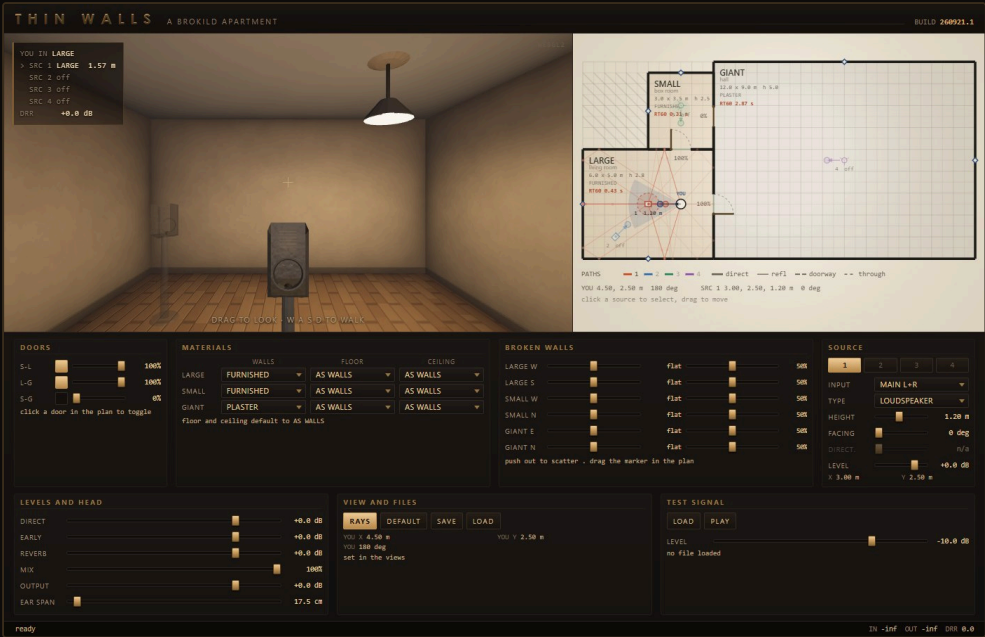


The flat. Living room bottom left, box room above it, hall to the east. The dark block top left is outside the apartment: nothing can stand there.

THE FIRST MINUTE

Put it on a track. Drag the loudspeaker glyph in the plan away from the head and back. Then click the door between the living room and the hall to shut it, and drag the source through into the hall. That is the whole instrument in three gestures.

TWO VIEWS AND A ROW OF CONTROLS



Left: the view from where you are standing. **Right:** the plan. **Below:** doors, materials, the selected source, levels and the test signal. **Foot:** input, output and the direct-to-reverberant ratio.

EVERY CONTROL EXPLAINS ITSELF

Hover any slider, button or menu and a panel appears naming it and saying what it is for. Nothing in this handbook is hidden from the instrument itself; the handbook is here for the things that need a picture or a table.

NOTHING IS MODAL

The two views and the controls are always live at once. A drag in the plan moves the 3D view as it happens, and walking in the 3D view moves the glyph in the plan. Both write the same host parameters, so both are recorded by your DAW's automation.

RAYS

The **Rays** button draws the engine's own paths on the plan: direct, first and second order reflections, doorway portals, and sound through a shut leaf. It is the fastest way to see why a move changed the sound.

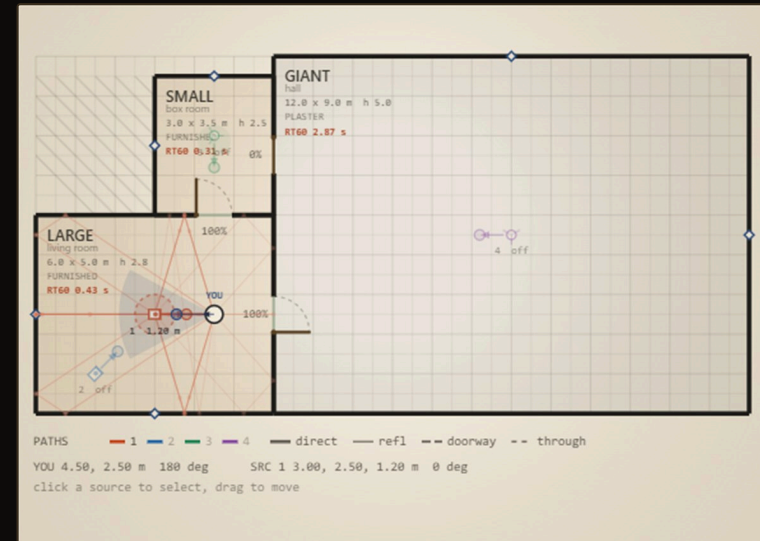
THE SEVEN GROUPS

Doors	one aperture each, and a button for the two ends	Broken walls	split a wall and push the break out, per room	View and files	rays, defaults, save and load, and where you are standing
Materials	nine choices — walls, floor and ceiling of each room	Levels and head	the three trims, mix, output and ear spacing	Test signal	a file of your own, looped into the input
Source	everything about whichever of the four is selected				

WALK, LOOK, DRAG



The 3D view is **you**. The camera is the listener's head, at 1.65 m. Drag inside it to look around; **W A S D** to walk. Walls stop you, and so does a door less than half open.



The plan is the **architect's drawing**. Drag any source or the head. Turn either by its heading handle. Click a source to select it for the controls below; click a door to toggle it, or drag along it to set it part open.

TURNING YOUR HEAD

Facing is not decoration. The binaural rendering is driven by the direction each arrival comes from **relative to your head**, so turning on the spot moves every source around you — including the ones you cannot see, behind a shut door.

WHERE YOU MAY STAND

Anywhere inside the three rooms. The area north-west of the living room is outside the flat and is solid; the plan will not let you drag there, and if automation sends you there the engine puts you back in the nearest room.

TAKE HOLD OF THE HANDLE

Every door in the 3D view has a handle. Walk up to one — within **2.5 m**, in front of you and not through a wall — and it lights up as the pointer crosses it. **Click** to slam it or throw it wide; **drag** to swing it to anywhere in between. Grabbing a handle is not a look, so the room does not spin while you do it.

THE KEYS ALWAYS WIN

W A S D and the arrow keys walk and turn **even when a menu or a fader has the focus**. Set a material, then walk — the key never reaches the menu behind it. A key typed into a text field is still the field's.

THREE ROOMS, THREE DOORS

Fixed, and deliberately so: every room is a different size and every pair of rooms has a door, so every “behind that one” case exists, including the two-door route through the third room.

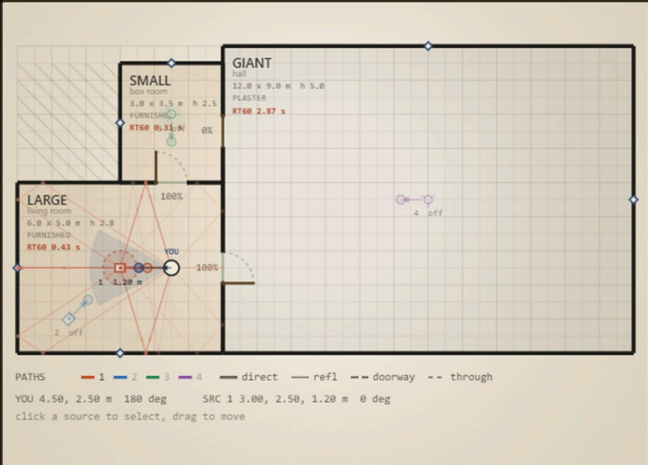
ROOMS

ROOM	SIZE (M)	HEIGHT	VOLUME WHAT IT IS
LARGE	6.0 × 5.0	2.8	84 m ³ the living room
SMALL	3.0 × 3.5	2.5	26 m ³ the box room
GIANT	12.0 × 9.0	5.0	540 m ³ the hall

DOORS

DOOR	BETWEEN	HINGED	DEFAULT
SMALL – LARGE	box room and living room	west side	open
LARGE – GIANT	living room and hall	south side	open
SMALL – GIANT	box room and hall	north side	shut

Every door is 0.90 m wide and 2.05 m high. Your ears are at 1.65 m; a source can stand between 0.2 and 2.4 m off the floor.



Origin is the south-west corner of the living room. X runs east, Y runs north. A facing of 0° is east and the angle increases anti-clockwise, so 90° is north.

THE TWO-DOOR ROUTE

Shut the living room door and the hall is not silent: it still reaches the box room through its own door, and from there back to you. Opening both doors of that route brings the sound up by **18 dB**. The path bends twice, and it is drawn.

FOUR OF THEM, AND TWO KINDS

Each source has its own position, height, facing, type, level and input. The numbered tabs pick which one the controls belong to; clicking a source in the plan picks it too.



LOUDSPEAKER — a modelled two-way monitor: a 6.5" woofer below 2.2 kHz, a 1" tweeter above. Nearly omnidirectional at 125 Hz, beaming with frequency in front, and about **22 dB down straight behind it at 4 kHz**.



PURE — a point source with no box, on a stand. **DIRECTIVITY** runs from omnidirectional to a cardioid and is the same at every frequency. The **HEIGHT** control is the stand.

FACING MATTERS

Turning a loudspeaker away from you takes **21.5 dB** off 4 kHz and only **5.5 dB** off 250 Hz. That is what a speaker pointed at the wall sounds like, and it is why aiming one into the hall and standing behind it is a usable effect rather than just a volume drop.

A DIRECTIONAL SOURCE DRIVES THE ROOM LESS

The room's tail is fed by the source's radiated **power**, not by what is aimed at you. Narrowing the pattern removes power, so a cardioid excites the room **4.8 dB** less than an omni at the same on-axis level. Point a speaker away and the room gets brighter relative to it, not darker.

HEIGHT

0.2 to 2.4 m. Low is floor-coupled and boomy, ear height is the most direct, high throws more into the ceiling. It also changes which ear is nearer, which is audible inside about a metre.

FEEDING THE FOUR

Every source has an **INPUT** menu with seven positions. Its first is **OFF**: the source makes no sound, costs nothing, and stays in the views as a ghost so you can place it before you use it. The other six take a channel from one of two stereo buses.

SETTING	WHAT FEEDS THE SOURCE
OFF	nothing — silent, ghosted, still placeable
MAIN L / R	one channel of the track the plugin is on
MAIN L+R	both, summed
AUX L / R	one channel of the second input bus
AUX L+R	both, summed

The obvious use of MAIN L and MAIN R is a stereo pair: two loudspeakers, one fed each channel, standing where you would put them in a room. The obvious use of AUX is a second instrument in a different room, sharing one apartment.

ROUTING THE AUX BUS

The plugin declares a second stereo input bus named **Aux In**, disabled until the host enables it. Most DAWs call this a **sidechain**. In Ableton Live, open the plugin's device title bar, switch on **Sidechain** and pick the track you want to send; in Reaper, add channels 3–4 to the track and pin the other track's send into them; in Cubase and Studio One, use the plugin's own side-chain button and route a send to it.

The panel greys the AUX entries out until the host has actually enabled the bus, so a source can never be pointed at silence without saying so.



The **source panel**. The tabs across the top pick which of the four you are editing.

FOUR SOURCES, ONE APARTMENT

Sources in the same room share that room's tail, which is correct: one room has one reverberant field however many things are shouting into it. Sources in different rooms drive their own rooms and reach you through the doors.

One source costs about **11 %** of a core at 48 kHz. Four, with every door open, cost **28 %**. Switching a source to OFF gives its share straight back.

WHAT THE WALLS ARE MADE OF

Five surface treatments, and any of them on any surface of any room. These are published octave-band absorption coefficients, not voicing presets, and the reverberation time follows from them.

THE FIVE

MATERIAL WHAT IT IS

ABSORBING	heavy drapes, thick carpet, soft furniture
FURNISHED	a lived-in living room: plaster, carpet, curtains
PLASTER	bare plaster on brick, wooden floor
TILED	glazed tiles, a little glass and wood
STUDIO	treated: broadband absorbers, bass traps and diffusers — page 10

Absorption rises with frequency on the first four, which is why a live room here gets darker as it decays rather than simply longer. ABSORBING takes 80 % at 4 kHz and only 20 % at 125 Hz, so a dead room still has a bloom in the bottom octave — as real ones do. STUDIO is the one that does not, and that is its whole point.

AN OPEN DOOR IS A HOLE

Sound that leaves through a doorway does not come back, so the engine counts an open door as a perfect absorber. The box room in bare plaster rings for **1.32 s** at 1 kHz with its doors shut and **0.58 s** with both wide open. You can hear a room change size by opening a door into it.

REVERBERATION TIME, DOORS SHUT

Seconds, from Eyring's formula with ISO 9613-1 air absorption — the same arithmetic the engine runs at every change. STUDIO is on [page 10](#).

ROOM	MATERIAL	125 HZ	500 HZ	1 KHZ	4 KHZ
LARGE	ABSORBING	0.51	0.13	0.09	0.07
LARGE	FURNISHED	1.07	0.65	0.51	0.34
LARGE	PLASTER	3.57	2.16	1.77	1.17
LARGE	TILED	5.24	5.18	3.42	2.07
SMALL	ABSORBING	0.38	0.10	0.07	0.06
SMALL	FURNISHED	0.78	0.48	0.38	0.25
SMALL	PLASTER	2.50	1.59	1.32	0.90
SMALL	TILED	3.59	3.83	2.57	1.64
GIANT	ABSORBING	0.92	0.23	0.15	0.13
GIANT	FURNISHED	1.94	1.16	0.90	0.58
GIANT	PLASTER	6.58	3.80	3.06	1.88
GIANT	TILED	9.78	8.90	5.75	3.06

Each door slider is an aperture, continuous from shut to wide open; the button beside it does the two ends. Clicking a door in the plan does the same thing, dragging along one sets it part open, and in the 3D view you can simply take hold of its handle.

WALLS, FLOOR AND CEILING, EACH ITS OWN

Nine choices, three for each room. Floor and ceiling start at AS WALLS, so a room you never touch is exactly the room it always was.

A real room is not made of one thing. Between a source standing on the floor and a listener standing on the floor, the **first two reflections to arrive are the one off the floor and the one off the ceiling** — they are the shortest detours there are. They arrive within a few milliseconds of the direct sound, which is exactly the window the ear uses to decide how big a room is and where in it you are.

So the floor and the ceiling are worth their own choices, and they are the two a real studio treats first.

A CARPET IS THE MOST LOPSIDED ABSORBER IN THE FLAT

Thick carpet takes almost nothing below 250 Hz and a great deal above 2 kHz. Setting **floor** to ABSORBING therefore kills the floor bounce and the flutter with it while leaving the bottom of the room exactly where it was — which is what laying a rug in a hard room really does, and why it is the first thing anyone tries.

A CLOUD OVER THE DESK

Ceiling set to ABSORBING is the cloud every control room has above the listening position. It is the opposite trade to a carpet: it takes the vertical flutter out and leaves the room's width and depth ringing.

MATERIALS

	WALLS	FLOOR	CEILING
LARGE	PLASTER	ABSORBING	ABSORBING
SMALL	FURNISHED	AS WALLS	AS WALLS
GIANT	STUDIO	AS WALLS	AS WALLS

floor and ceiling default to AS WALLS

Nine selectors. A row per room, a column per surface. AS WALLS is not a fifth material, it is a pointer: that surface follows whatever the walls are set to.

WHAT IT IS WORTH

The living room in bare plaster rings for **2.71 s** at 250 Hz and **1.77 s** at 1 kHz. Leave the walls in plaster and put a carpet down and a cloud up — two menu changes — and the same room measures:

SURFACES	250 HZ	1 KHZ	4 KHZ
plaster throughout	2.71	1.77	1.17
+ carpet and cloud	0.39	0.20	0.17

Read off the panel's own readout with the engine running. Two of the six surfaces carry most of the room.

STUDIO: TREATED, AND STILL LIVE

A room whose decay does not move with frequency, and whose treatment is diffusion rather than absorption.

The other four materials are measured surfaces. STUDIO is not: its absorption was **solved backwards from Eyring's formula** for the one property a control room is built to have — a reverberation time that is the same at the bottom of the band as at the top.

What comes out is **constant absorption near a quarter**, with a little relief at the very top to pay for the air absorption that takes the top octave anyway. That is not a coincidence: broadband absorbers plus bass traps are exactly what it takes to flatten a real room, and this is the coefficient curve they add up to.

ROOM AS A STUDIO	125 HZ	500 HZ	1 KHZ	4 KHZ	SPREAD
LARGE · living room	0.39	0.40	0.40	0.41	1.04×
SMALL · box room	0.29	0.29	0.30	0.31	1.06×
GIANT · hall	0.71	0.71	0.71	0.71	1.00×
GIANT · the same hall, TILED	9.78	8.90	5.75	3.06	3.09×

Spread is the ratio of the longest of 250 Hz, 1 kHz and 4 kHz to the shortest, measured on the rendered audio. The hall as a studio is **1.00×**: flat to the digit.

DIFFUSION, NOT ABSORPTION

The other four materials return a reflection in the specular direction and take a bite out of it. A studio's diffusers do something quite different: they **keep the energy and throw away the direction**.

So what comes back off a treated wall is still in the room — it just no longer arrives as a single sharp image, and it no longer bounces between two parallel walls in step with itself. Measured: the specular reflections carry **17.6 %** of what the same room in plaster returns, and the rest is scattered into the room rather than absorbed out of it.

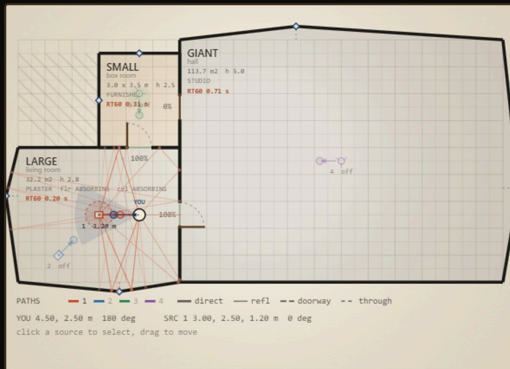
WHICH IS WHY IT IS NOT DEAD

A hall that rings for 0.71 s is a live room, not an anechoic chamber. You can still hear that it is big. What you cannot hear is a flutter echo, or the bottom end arriving late: the early response of a centred source measures **1.1 dB** uneven across the middle of the band, where the same room in plaster measures **2.4 dB**.

STUDIO also splays its walls very slightly, which is the same trick the next page gives you by hand — and the two add up.

THE ROOMS ARE NOT SHOEBOXES ANY MORE

Two walls of every room can be split at a movable point and that point pushed out, so the room stops being rectangular. Drag the diamond in the plan, or use the two faders.



Four walls broken. The diamond on each is the break: drag it along the wall, or out of it. Area, volume and reverberation time all follow — the hall's floor goes from 108.0 to **113.7 m²**, and its own decay from 1.766 s to **1.801 s**.

PUSH IT OUT

SPLAY

Two panels that are no longer parallel to the wall facing them cannot sustain a flutter between them, and they spread a reflection over a range of angles instead of returning it whole. Off centre is worth more than centred: two panels of different lengths scatter over a wider band than two of the same length.

DO NOT PULL IT IN

CONCAVE FOCUSES

Pushed the other way the wall bends **into** the room, and a concave surface gathers sound to a point instead of spreading it. That is worse than leaving the wall flat — real rooms are splayed outward, never inward. The plan says so when you do it. It is offered because it is audible and instructive, not because it is good.



Two faders per wall. How far the break is pushed (± 0.6 m, **flat** at the centre) and where along the wall it sits. The two walls offered are always the ones with no doorway, so a break can never collide with a door.

WHAT IT REALLY DOES

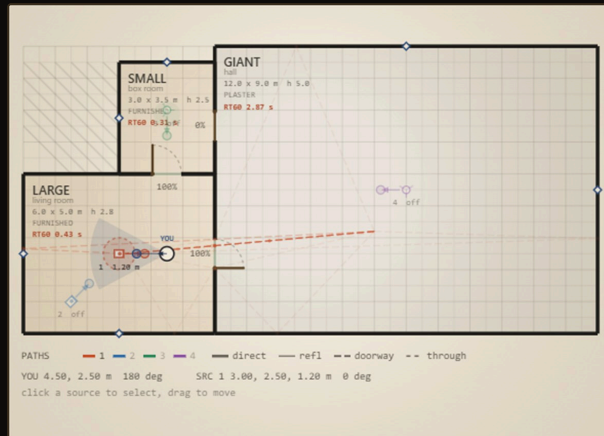
HONESTLY

In this model the splay acts on the **early reflections**, not on the room's standing waves: the late field is statistical rather than modal. Above the Schroeder frequency — about 72 Hz in the hall as a studio — that is where splay does its audible work in a real room too. Below it, absorption is what helps.

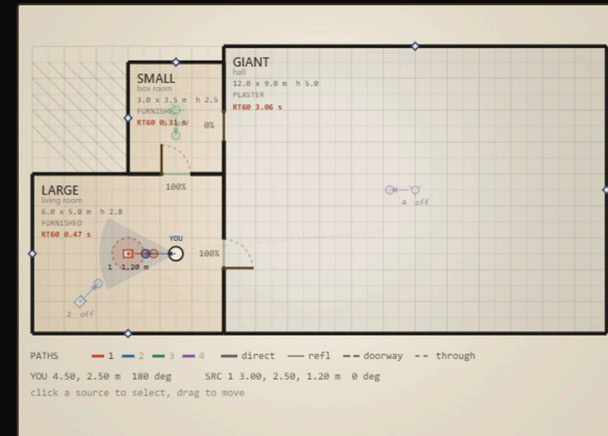
A room that is not a box needs a different search. A room is now a list of planar surfaces and a path a sequence of them, traced back from the listener and validated at every step; a square room still takes the exact arithmetic. A wall broken by **two millimetres** is the same room, and the general search reproduces the exact one to **0.09 dB** overall and **0.20 dB** over the first 50 ms.

THREE DIFFERENT WAYS IN

A sound in the next room reaches you by several routes at once, and they do not sound alike. This is the part of the instrument the name is about.



Door open. The path bends at the edge of the opening and the sound is localised **at the doorway**, not at the source — exactly as it is in a real flat.



Door shut. What is left comes through the leaf and through the party wall, radiated from the point of each nearest the straight line, so it still comes from the right direction.

ROUND THE FRAME

DIFFRACTION

Inside the open strip it is line of sight. Outside it, the path bends at the nearest point of the opening and pays Maekawa's attenuation, which costs high frequencies far more than low ones — the bent path loses **6 dB more at 4 kHz** than the clear one. The edge fades in rather than switching on, so walking past a doorway never clicks.

THROUGH THE LEAF

A LIGHT INTERIOR DOOR

A shut door is not a wall. It passes the mass law of a light leaf — **14 dB** of loss at 125 Hz rising to **32 dB** at 8 kHz. Shutting it drops 1 kHz by **20 dB** and darkens what is left: 4 kHz loses **8 dB more** than 250 Hz. That is the sound of a party next door.

THROUGH THE WALL

THE PARTY WALL

Heavier: **30 dB** at 125 Hz to **53 dB** at 8 kHz. It is usually below the door, but with every door shut it is what you are left with — measured **24 dB** below the same source at the same distance in your own room. Bass, and almost nothing else.

A door part open sits between the two, continuously: the engine does not crossfade a shut sound into an open one. The opening is a real strip of a real width, and the diffraction, the leaf and the room's own absorption all follow from it.

WHY THERE IS NO REVERB AMOUNT

Each room runs its own sixteen-line reverberator, with delays taken from that room's mean free path and a decay per octave band from its materials, its open doors and the air in it. The rooms are coupled through the doors and walls with the energy balance of coupled-room theory, so the hall's long tail genuinely arrives in the living room and gives you the double slope a real flat has.

Inside every one of those lines sits a chain of **allpass diffusers**. An allpass has unity gain at every frequency, so it cannot change the decay — but its length counts towards the loop, which decides whether the network's resonances are close enough together to hear as a room rather than as ringing, and it splits each echo into a train, which decides whether the tail has grain. A perfectly diffuse field has a spectral roughness of **5.57 dB**; this one measures **5.6 to 6.6**, where before the diffusers it measured **6.1 to 10.7**.

What that field is **worth**, relative to the direct sound, is not a taste decision. For a diffuse field it is fixed by the room's total absorption and the engine calibrates it to exactly that, so the direct-to-reverberant ratio — the strongest cue your ears have for distance — comes out right by construction. Walk twice as far away and it falls by **6.2 to 7.0 dB**; walk in to half the distance and it rises by **3.6 to 5.5**. At the critical distance, where direct and reverberant are equal, it reads between **-1.3 and -0.4 dB** against the **+0.9 dB** the head's own frontal gain accounts for — for every one of the five materials.

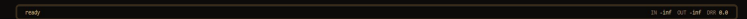
That is why there is no “reverb amount”: turning one up would mean lying about how far away you were standing. What you get instead is three trims — DIRECT, EARLY and REVERB — each at 0 dB, for when you want the lie on purpose.

THE DOUBLE SLOPE

A flat does not have one reverberation time, it has as many as it has rooms. Stand in the box room with it lined in drapes and the hall in tiles: its own field is **16 dB down inside 40 ms** and gone, and then the hall's answer comes back through the doorway at the hall's own **5.8 seconds**. The decay falls off a cliff and then keeps going — which is what a small dead room inside a live building does, and what a single reverb cannot do at all. Shut the door and that second slope goes with it.



Top left of the 3D view: which room you are in, and for each of the four sources its room and its distance from you. An OFF source still gets a line, marked, because knowing which one is silent is half of what the readout is for.



Along the foot: input and output level, and the direct-to-reverberant ratio in dB. Positive means you are inside the critical distance.

READING THE DRR

It counts every arrival that took the shortest route — including through a doorway or a shut leaf — as “direct”. That is the useful reading rather than the textbook one: it tells you how much of what you hear is the sound itself and how much is the building.

SIX CONTROLS AT THE END OF THE CHAIN

DIRECT

–24 TO +6 DB, DEFAULT 0

The straight line of sight. Pull it down and the room is all you have left — a way of moving a source further away without moving it.

REVERB

–24 TO +6 DB, DEFAULT 0

The coupled tails. The hall's arrives through whatever the doors are doing, so this and the doors do different jobs.

OUTPUT

–24 TO +6 DB, DEFAULT 0

The last thing in the chain. Two sources in one room and four in three can add up to more than one did, so there is somewhere to give it back.

EARLY

–24 TO +6 DB, DEFAULT 0

The wall bounces and the doorway paths. This is the part that tells you the size and the shape of the room you are in.

MIX

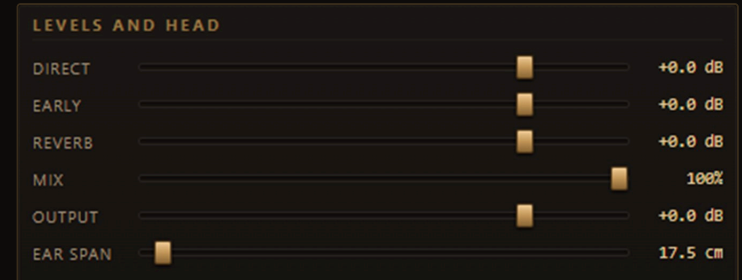
0 TO 100 %, DEFAULT 100

Dry against placed. At 100 you are entirely in the apartment. At 0 the input passes through untouched, exactly.

EAR SPACING

15 TO 100 CM, DEFAULT 17.5

How far apart your two ears sit. 17.5 cm is a head; a metre is a spaced pair of microphones, and the image opens out accordingly.



Levels and head, with the view and file buttons below: Rays, Default, Save, Load, and a readout of where you are standing.

WHAT EAR SPACING REALLY DOES

It stretches the interaural time difference and the two ear distances, leaving the pinna colouration alone — so it is a head on a wider neck, not a bigger head. Past about 25 cm it stops being a person and becomes a spaced microphone pair, which is a perfectly good thing to be: it widens the image without the phase trouble a stereo widener brings.

THE TEST SIGNAL AND THE STANDALONE

Walking around a room is not something you can do with one hand on a transport bar, so the plugin can play its own sound.

Load picks a WAV, AIFF, FLAC or MP3 off disk. **Play** loops it into the **MAIN** input — both channels — so every source fed from MAIN plays it, and you are free to spend both hands on the views. Sources fed from AUX are untouched, so you can audition one thing while a second track keeps running in another room.

Stop hands MAIN back to the host, immediately. The test signal is not a host parameter and is not saved in a patch: it is an audition tool, not part of the sound.

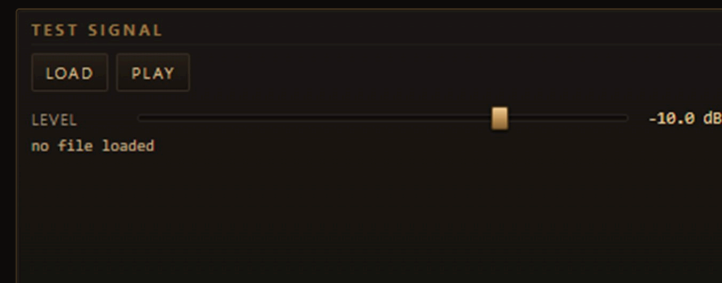
The same controls are in the VST3 and in the standalone. The standalone needs no host at all — open it, load a loop, and walk the flat.

PATCHES

Save and **Load** write and read a JSON file in `DOCUMENTS\BROKILD PATCHES\THIN WALLS`. A patch holds every surface, every broken wall, the doors, all four sources and the levels — the whole arrangement. **Default** puts every parameter back; the floor plan itself never changes.

WHAT TO WALK AROUND TO

Something broadband and rhythmic. A shut door takes **20 dB** off 1 kHz and a good deal more off the top, so a loop with transients and some air in it tells you where you are far faster than a pad does — a drum loop, a spoken word, a strummed chord. Sustained, low-passed material is exactly the hardest case for the ear to place, and the flat will sound less impressive than it is.



The **test signal group**. The line underneath names the loaded file and its length.

AUTOMATION

All **65** parameters are host parameters, including every source position, every surface, every broken wall and your own position. Walking is recorded as automation, so a move you made by hand can be replayed exactly — and a source can be given a path by drawing one in your DAW.

A source moving at 2 m/s on a held tone keeps everything above 6 kHz **84 dB down**, and a wall moved under a sounding note **92 dB down**: the geometry changes continuously and nothing clicks.

WHAT WAS MEASURED

The bench renders real audio and measures it: 90 checks, all clear, beside 73 that drive the panel. These are their numbers, not estimates.

ITEM	MEASURED	ITEM	
Interaural delay at 90°	−708 μs (Woodworth's formula gives 666)	Format	Windows VST3 (64-bit), plus a standalone application
Front against back	5.0 dB apart in spectral shape at 4 and 8 kHz, 1.0 dB in level — the pinna	Type	Effect — stereo in, stereo out, plus a second stereo input bus ("Aux In", the host's sidechain)
Level with distance	6.1 dB per doubling — and the interaural level difference grows 4.6 dB by 0.3 m, out of the geometry	Parameters	65, all automatable
Direct path	within a tenth of a decibel of the physics from 10 Hz to 16 kHz, wherever a source stands	Materials	Five, chosen per surface — walls, floor and ceiling of each room, nine choices. Two walls of every room can be broken and pushed ±0.6 m
Air absorption	8 kHz costs 0.73 dB more than 250 Hz over 11 m (ISO 9613-1: 0.78)	Head	MIT KEMAR compact set, 368 directions, diffuse-field equalised, converted to minimum phase with a separate onset delay, resampled to the host rate and interpolated on the grid
Reverberation time	against Eyring in all fifteen room and material cases; the tail's spectral roughness 5.6 to 6.6 dB, against the 5.57 dB of a perfectly diffuse field	Reflections	image sources to second order; a general surface-sequence search once a wall is broken
Direct-to-reverberant	−1.3 to −0.4 dB at the critical distance against the head's own +0.9; +3.6 / −7.0 at half and double	Reverberator	one 16-line feedback delay network per room, allpass diffusers inside every line, coupled through the doors and walls
Shutting a door	−23 dB at 1 kHz, and 4 kHz loses 12 dB more than 250 Hz	Delay reader	16-tap windowed sinc, so a source may stand anywhere without losing the top octave
Party wall and shut door	31 dB below the same distance in your own room	CPU	one source about 17 % of a core at 48 kHz; four with every door open, 40 %
Moving a source	2 m/s on a pure tone: everything above 6 kHz stays 84 dB down	Sample rates	any; bench-tested at 44.1 and 96 kHz
MIX at 0	bit-exact passthrough	Hosts	any VST3 host — developed against Ableton Live 12
		Price	free. No account, no registration, no telemetry.

The head data is the MIT KEMAR HRTF measurements of **W. G. Gardner and K. D. Martin (1994)**, MIT Media Lab, used with thanks. Absorption coefficients from the standard tables (Kuttruff; Vorländer). Diffraction after Maekawa; air absorption after ISO 9613-1.

HONEST LIMITS

WHAT IT IS NOT

The floor plan is fixed. One apartment, three rooms, three doors. Two walls of each room can be broken and pushed, so the rooms are not stuck as boxes — but where the rooms are, and which doors join them, you cannot change. A second plan would be a different instrument.

Reflections are specular and go to second order. Only STUDIO scatters, and there is no diffraction round the edges of the reflections themselves. Beyond the second bounce the room's own reverberator takes over, which is where that energy belongs anyway.

Two sources in one room share its field. That is correct — a room has one reverberant field — but the share each hands it is taken at 1 kHz rather than per band.

Breaking a wall moves the early reflections, not the modes. The late field here is statistical, so splaying a wall does not redistribute a room's standing waves the way it would in a modal model. Above the Schroeder frequency that is where splay works in a real room too; below it, absorption is what helps.

The DRR readout is the useful reading, not the textbook one. It counts an arrival that came through a doorway or a shut leaf as direct, because that is the sound rather than the building. And the test signal **replaces** the MAIN input while it plays, rather than mixing with it; AUX carries on regardless.

None of these is a placeholder. They are the edges of what was built, written down so you know where they are.

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MIND THE NEIGHBOURS

